

Part 1: Capturing HTTP Traffic

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Task 2: Filter HTTP packets and analyze them

Step 3: HTTP Request and Response Observations:

- **HTTP Request Details:**
 - **HTTP Method:** GET
 - **URL/Host:** http://neverssl.com/
 - **User-Agent:** Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)

The image displays a Wireshark packet capture of HTTP traffic. The packet list shows a GET request for http://neverssl.com/. The packet details pane shows the structure of the HTTP request, including the User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/152.0.0.0 Safari/537.36. The packet bytes pane shows the raw data of the request.

No.	Time	Source	Destination	Protocol	Length	Info
390	15.482873	2a02:cb80:4029:f0e...	2600:1f13:37c:1400...	HTTP	519	GET / HTTP/1.1
398	15.757581	2600:1f13:37c:1400...	2a02:cb80:4029:f0e...	HTTP	1071	HTTP/1.1 200 OK (text/html)
542	18.867633	192.168.8.53	34.223.124.45	HTTP	556	GET /online HTTP/1.1
549	19.166956	34.223.124.45	192.168.8.53	HTTP	639	HTTP/1.1 301 Moved Permanently (text/html)
551	19.172773	192.168.8.53	34.223.124.45	HTTP	557	GET /online/ HTTP/1.1
564	19.486476	34.223.124.45	192.168.8.53	HTTP	297	HTTP/1.1 200 OK (text/html)
567	19.477170	192.168.8.53	34.223.124.45	HTTP	487	GET /favicon.ico HTTP/1.1
571	19.794172	34.223.124.45	192.168.8.53	HTTP	482	HTTP/1.1 200 OK (PNG)

Frame 390: Packet, 519 bytes on wire (4152 bits), 519 bytes captured (4152 bits) on interface 0: Ethernet II, Src: Apple_00:27:73 (a4:83:e7:60:27:73), Dst: 02:b0:33:93:70:c0 (02:b0:33:93:70:c0), Internet Protocol Version 6, Src: 2a02:cb80:4029:f0e5:f78:d75e:cdfe, Dst: 2600:1f13:37c:1400:1f13:37c:1400:1f13, Transmission Control Protocol, Src Port: 49687, Dst Port: 80, Seq: 1, Ack: 1, Len: 433, Hypertext Transfer Protocol

0000 02 b0 33 93 70 c0 a4 83 e7 60 27 73 86 dd 60 05 ...3.p... 's...
0010 06 00 01 d1 06 40 2a 02 cb 80 40 29 f0 ed 50 f5 ...@...@)P...
0020 0f 78 d7 5e cd fe 26 00 1f 13 03 7c 14 00 ba 21 ...x^&...|...!
0030 71 65 5f c7 73 6e c2 17 00 50 6d 13 f6 9d 5a ad ...qe_s...Pm...Z...
0040 17 b8 08 18 08 0c dd 94 00 00 01 01 08 0a 7f ce ...qGE T / HTTP
0050 00 ec 19 c9 f7 71 47 45 54 20 2f 28 48 54 54 50 .../1.1- Ho st: neve
0060 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 6e 65 76 65 ...rssl.com - Conne
0070 72 73 73 6c 2e 63 6f 6d 0d 0a 43 6f 6e 6e 65 63 ...tion: ke ep-alive
0080 74 69 6f 6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65 ...Upgrad e-Insecu
0090 0d 0a 55 70 67 72 61 64 65 2d 49 6e 73 65 63 75 ...re-Reqe sts: 1-
00a0 72 65 2d 52 65 71 75 65 73 74 73 3a 20 31 0d 0a ...User-Age nt: Mozi
00b0 55 73 65 72 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69 ...lla/5.0 (Macinto
00c0 6c 6c 61 2f 35 2e 30 20 28 4d 61 63 69 6e 74 6f ...sh; Inte l Mac OS
00d0 73 68 3b 20 49 6e 74 65 6c 20 4d 61 63 20 4f 53 ...X 10_15_7) Appl
00e0 20 58 20 31 30 5f 31 35 5f 37 29 20 41 70 70 6c ...eWebKit/ 537.36 (
00f0 65 57 65 62 4b 69 74 2f 35 33 37 2e 33 36 20 28 ...KHTML, l ike Geck
0100 4b 48 54 4d 4c 2c 20 6c 69 6b 65 20 47 65 63 6b ...o) Chrom e/152.0.
0110 6f 29 20 43 68 72 6f 6d 65 2f 31 35 32 2e 30 2e ...0.0 Safa ri/537.3
0120 30 2e 30 20 53 61 66 61 72 69 2f 35 33 37 2e 33 ...6-Accep t: text/
0130 36 0d 0a 41 63 63 65 70 74 3a 20 74 65 78 74 2f ...html,app lication
0140 68 74 6d 6c 2c 61 70 70 6c 69 63 61 74 69 6f 6e .../html/x ml,appli
0150 2f 78 68 74 6d 6c 2b 78 6d 6c 2c 61 70 70 6c 69 ...cation/x ml;q=0.9
0160 63 61 74 69 6f 6e 2f 78 6d 6c 3b 71 3d 30 2e 39 ... ,image/a vif,imag
0170 2c 69 6d 61 67 65 2f 61 76 69 66 2c 69 6d 61 67 ...e/webp,i mage/apn
0180 65 2f 77 65 62 70 2c 69 6d 61 67 65 2f 61 70 6e ...g,*/*;q= 0.8,appl
0190 67 2c 2a 2f 2a 3b 71 3d 30 2e 38 2c 61 70 70 6c

HTTP Response Details:

- Response Code: 200 OK
- Content-Type: text/html
- Server: Apache/2.4.68

http

No.	Time	Source	Destination	Protocol	Length	Info
390	15.482873	2a02:cb80:4029:f0e...	2600:1f13:37c:1400...	HTTP	519	GET / HTTP/1.1
398	15.757581	2600:1f13:37c:1400...	2a02:cb80:4029:f0e...	HTTP	1071	HTTP/1.1 200 OK (text/html)
542	18.867633	192.168.8.53	34.223.124.45	HTTP	556	GET /online HTTP/1.1
549	19.166956	34.223.124.45	192.168.8.53	HTTP	639	HTTP/1.1 301 Moved Permanently (text/html)
551	19.172773	192.168.8.53	34.223.124.45	HTTP	557	GET /online/ HTTP/1.1
564	19.486476	34.223.124.45	192.168.8.53	HTTP	297	HTTP/1.1 200 OK (text/html)
567	19.477170	192.168.8.53	34.223.124.45	HTTP	487	GET /favicon.ico HTTP/1.1
571	19.794172	34.223.124.45	192.168.8.53	HTTP	482	HTTP/1.1 200 OK (PNG)

> Frame 571: Packet, 482 bytes on wire (3856 bits), 482 bytes captured (3856 bits) on interface 0
> Ethernet II, Src: 02:b0:33:93:70:c0 (02:b0:33:93:70:c0), Dst: Apple_60:27:73 (a4:83:e7:60:27:73)
> Internet Protocol Version 4, Src: 34.223.124.45, Dst: 192.168.8.53
> Transmission Control Protocol, Src Port: 80, Dst Port: 49697, Seq: 2093, Ack: 1403, Len: 482
Hypertext Transfer Protocol
 > HTTP/1.1 200 OK\r\n Date: Wed, 16 Sep 2026 22:50:16 GMT\r\n Server: Apache/2.4.68 (Ubuntu)\r\n Last-Modified: Wed, 29 Jun 2022 00:26:43 GMT\r\n ETag: "7c-5e28b351ffe27"\r\n Accept-Ranges: bytes\r\n Content-Length: 124\r\n Keep-Alive: timeout=5, max=98\r\n Connection: Keep-Alive\r\n Content-Type: image/vnd.microsoft.icon\r\n \r\n [Request in frame: 567]
 [Time since request: 317.002000 milliseconds]
 [Request URI: /favicon.ico]
 [Full request URI: http://glowinglushfinerain.neverssl.com/favicon.ico]
 File Data: 124 bytes
 > Portable Network Graphics

0040	d3 1f 48 54 54 50 2f 31	2e 31 20 32 30 30 20 4f	HTTP/1.1 200 OK
0050	4b 0d 0a 44 61 74 65 3a	20 57 65 64 2c 20 31 36	Date: Wed, 16
0060	20 53 65 70 20 32 30 32	36 20 32 32 3a 35 30 3a	Sep 2026 22:50:
0070	31 36 20 47 4d 54 0d 0a	53 65 72 76 65 72 3a 20	16 GMT; Server:
0080	41 70 61 63 68 65 2f 32	2e 34 2e 36 38 20 28 29	Apache/2.4.68 (
0090	0d 0a 4c 61 73 74 2d 4d	6f 64 69 66 69 65 64 3a	Last-Modified:
00a0	20 57 65 64 2c 20 32 39	20 4a 75 6e 20 32 30 32	Wed, 29 Jun 202
00b0	32 20 30 30 3a 32 36 3a	34 33 20 47 4d 54 0d 0a	20:26:43 GMT;
00c0	45 54 61 67 3a 20 22 37	63 2d 35 65 32 38 62 33	ETag: "7c-5e28b3
00d0	35 31 66 66 65 32 37 22	0d 0a 41 63 63 65 70 74	51ffe27"; Accept
00e0	2d 52 61 6e 67 65 73 3a	20 62 79 74 65 73 0d 0a	Content-Length:
00f0	43 6f 6e 74 65 6e 74 2d	4c 65 6e 67 74 68 3a 20	124; Keep-Alive:
0100	31 32 34 0d 0a 4b 65 65	70 2d 41 6c 69 76 65 3a	timeout=5, max=
0110	20 74 69 6d 65 6f 75 74	3d 35 2c 20 6d 61 78 3d	98; Connection:
0120	39 38 0d 0a 43 6f 6e 6e	65 63 74 69 6f 6e 3a 20	Keep-Alive: Cont
0130	4b 65 65 70 2d 41 6c 69	76 65 0d 0a 43 6f 6e 74	ent-Type: image/
0140	65 6e 74 2d 54 79 70 65	3a 20 69 6d 61 67 65 2f	vnd.microsoft.i
0150	76 6e 64 2e 6d 69 63 72	6f 73 6f 66 74 2e 69 63	on; PNG
0160	6f 6e 0d 0a 0d 0a 89 50	4e 47 0d 0a 1a 0a 00 00	IHDR
0170	00 0d 49 48 44 52 00 00	01 30 00 00 01 30 01 03	5
0180	00 00 00 0a 35 bf b0 00	00 00 05 50 4c 54 45 ff	PLTE
0190	ff ff 42 c0 fd 2e 2a d3	3f 00 00 00 31 49 44 41	B; ?
01a0	54 68 de ed ca 31 0d 00	00 08 03 30 fc 9b 06 0b	Th
01b0	24 7b 76 b4 77 67 5f 34	4d d3 34 4d d3 34 4d d3	\$(v w g 4 M 4M
01c0	34 4d d3 b4 a8 01 00 00	00 00 00 00 f4 39 04 0f	4M
01d0	01 ee 06 30 dc 50 00 00	00 00 49 45 4e 44 ae 42	0-P; IEND B

Hypertext Transfer Protocol (http), 292 bytes

Packets: 608 - Displayed: 8 (1.3%) - Dropped: 0 (0.0%)

Profile: Default

The figure displays a Wireshark packet capture interface. The top pane shows a list of network packets. Packet 564 is highlighted in red, indicating it is selected. The middle pane shows the details of this packet, which is a Transmission Control Protocol (TCP) segment. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
454	16.080488	192.168.8.53	34.223.124.45	TCP	78	49697 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=3791832406 TSecr=0 SACK_PERM
520	17.079886	192.168.8.53	34.223.124.45	TCP	78	[TCP Retransmission] 49697 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=3791833406 TSecr=0 S...
526	18.081254	192.168.8.53	34.223.124.45	TCP	78	[TCP Retransmission] 49697 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=3791834407 TSecr=0 S...
533	18.380876	34.223.124.45	192.168.8.53	TCP	74	80 → 49697 [FIN, ACK] Seq=1 Ack=1 Win=26847 Len=0 MSS=1400 SACK_PERM=1 TSval=3879028411 TSecr=379183440...
534	18.380883	192.168.8.53	34.223.124.45	TCP	66	49697 → 80 [ACK] Seq=1 Ack=1 Win=131840 Len=0 TSval=3791834787 TSecr=3791834407
542	18.867633	192.168.8.53	34.223.124.45	HTTP	556	GET /online HTTP/1.1
548	19.157519	34.223.124.45	192.168.8.53	TCP	66	80 → 49697 [ACK] Seq=1 Ack=491 Win=27136 Len=0 TSval=3879029254 TSecr=3791835194
549	19.166956	34.223.124.45	192.168.8.53	HTTP	639	HTTP/1.1 301 Moved Permanently (text/html)
550	19.167089	192.168.8.53	34.223.124.45	TCP	66	49697 → 80 [ACK] Seq=491 Ack=574 Win=131264 Len=0 TSval=3791835493 TSecr=3879029254
551	19.172773	192.168.8.53	34.223.124.45	HTTP	557	GET /online/ HTTP/1.1
563	19.486475	34.223.124.45	192.168.8.53	TCP	1354	80 → 49697 [ACK] Seq=574 Ack=982 Win=28160 Len=1288 TSval=3879029569 TSecr=3791835499 [TCP PDU reasse...
564	19.486476	34.223.124.45	192.168.8.53	HTTP	297	HTTP/1.1 200 OK (text/html)
565	19.486605	192.168.8.53	34.223.124.45	TCP	66	49697 → 80 [ACK] Seq=982 Ack=2093 Win=129728 Len=0 TSval=3791835813 TSecr=3879029569
567	19.477170	192.168.8.53	34.223.124.45	HTTP	487	GET /favicon.ico HTTP/1.1
571	19.794172	34.223.124.45	192.168.8.53	HTTP	482	HTTP/1.1 200 OK (PNG)
572	19.794301	192.168.8.53	34.223.124.45	TCP	66	49697 → 80 [ACK] Seq=1403 Ack=2509 Win=130624 Len=0 TSval=3791836252 TSecr=3879030008
600	24.961118	34.223.124.45	192.168.8.53	TCP	66	80 → 49697 [FIN, ACK] Seq=2509 Ack=1403 Win=29312 Len=0 TSval=3879035913 TSecr=3791836252
601	24.961249	192.168.8.53	34.223.124.45	TCP	66	49697 → 80 [ACK] Seq=1403 Ack=2510 Win=131072 Len=0 TSval=3791841419 TSecr=3879035013

Transmission Control Protocol, Src Port: 80, Dst Port: 49697, Seq: 1862, Ack: 982, Len: 0	
Source Port:	80
Destination Port:	49697
[Stream index: 15]	
[Stream Packet Number: 12]	
> [Conversation completeness: Complete, WITH_DATA (31)]	
[TCP Segment Len: 231]	
Sequence Number:	1862 (relative sequence number)
Sequence Number (raw):	340280886
[Next Sequence Number: 2093 (relative sequence number)]	
Acknowledgment Number:	982 (relative ack number)
Acknowledgment number (raw):	4215900036
1000 = Header length: 32 bytes (8)	
Flags:	0x018 (PSH, ACK)
Window: 220	
[Calculated window size: 28160]	
[Window size scaling factor: 128]	
Checksum:	0x5f23 [unverified]
[Checksum Status: Unverified]	
Urgent Pointer:	0
> Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps	
[Timestamps]	

Packet (297 bytes)	Reassembled TCP (1519 bytes)	Uncompressed entity body (2238 bytes)
0000 a4 83 e7 c0 2d 73 02 b0 33 93 70 c0 08 00 45 00 ...'s... 3 p... E-		
0010 01 1b 41 bc 40 0e 09 e0 67 36 22 d7 7c 2d c0 ab ...A @... 6'... [-...		
0020 08 35 00 50 c2 21 14 d7 2c a6 fb 3d 2b 74 80 18 ...5P!G...==+t... ..		
0030 00 dc 5f 23 00 00 01 01 08 0a e7 35 4b 41 e2 02 ...#.....5KA... ..		
0040 d1 66 75 be 36 44 2a 3b 25 05 4c 5e 09 80 13 a3 ...ku GDS; %L... ..		
0050 f2 79 8d 2c 70 55 82 85 16 87 86 a4 48 cf 8a cd ...y .xv... ..H... ..		
0060 bf 41 47 a2 9c 6a cb 38 33 42 1c 51 0d cf 3c 78 ...AG-j 8 3B 0<>.cx		
0070 6c e1 b1 bf 68 2c 68 91 87 f2 04 7d 29 49 21 a1 ...l..h.h...<{}I!..		
0080 f7 34 8e 39 14 70 25 95 0f 1e 4d 2a 3d 37 e3 54 ...4 9 p... M=-7 T	</	
0090 09 22 82 32 66 e4 79 25 36 04 56 ac 68 44 93 89 ...2F... 6 V..hd...		
00a0 69 76 e0 79 a2 b8 ae 65 9d 3b 10 0a 11 43 02 ...b b... 0a 11 43 02		
00b0 79 20 bb 1b 0c 0e 79 2f 4c aa 9d 2f 6a 6c 32 ...y...&L.../jL 2		
00c0 e0 73 c7 b3 c4 02 79 86 1a ab 4d ab 68 7c 6b ca ...y...&L...[k... ..		
00d0 76 6a 6e 4b 9d 67 29 3d 37 d3 e6 79 c8 31 51 4b ...vjkNg = 7...y:10K		
00e0 c2 27 38 1c 4f 28 30 78 82 76 27 31 0d b4 82 ...'8 0(8 ...v'1...		
00f0 46 ba 50 ab 86 94 79 9e 2b 08 4f 5d cf d7 d2 53 ...F-P...y...<0...S		
0100 3f 85 8f c5 3c 62 1b d5 7b 5d 14 e9 04 97 79 87 ...?...<b...<...<y...		
0110 3d 39 8e bf bf 19 dc bd f5 8d a0 86 0f bf b6 fc ...>...<...<...<...<y...		
0120 0b 05 79 44 e5 be 08 00 00		

3. TCP Connection Termination:

- Connection closure sequence using FIN, ACK flags to terminate the session safely.

The image shows a Wireshark packet capture of a TCP connection. The packet list at the top shows a sequence of packets, including a SYN packet (No. 454), a retransmission (No. 520), and several ACK packets. The packet details pane for packet 600 shows the following information:

- Transmission Control Protocol, Src Port: 80, Dst Port: 49697, Seq: 2509, Ack: 1403, Len: 0
- Source Port: 80
- Destination Port: 49697
- [Stream index: 15]
- [Stream Packet Number: 17]
- > [Conversation completeness: Complete, WITH_DATA (31)]
- [TCP Segment Len: 0]
- Sequence Number: 2509 (relative sequence number)
- Sequence Number (raw): 340209453
- [Next Sequence Number: 2510 (relative sequence number)]
- Acknowledgment Number: 1403 (relative ack number)
- Acknowledgment number (raw): 4215090457
- 1000 ... = Header Length: 32 bytes (8)
- > **Flags: 0x011 (FIN, ACK)**
- Window: 229
- [Calculated window size: 29312]
- [Window size scaling factor: 128]
- Checksum: 0xe195 [unverified]
- [Checksum Status: Unverified]
- Urgent Pointer: 0
- > Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
- > [Timestamps]

The packet bytes pane shows the raw data of the packet, including the FIN and ACK flags.

**** FOLLOW TCP STREAM ****

The image shows a Wireshark packet capture of an HTTP request and response. The packet list at the top shows a GET request (No. 572) and a 301 Moved Permanently response (No. 600). The packet details pane for packet 600 shows the following information:

- HTTP/1.1 301 Moved Permanently
- Date: Wed, 16 Sep 2026 22:50:15 GMT
- Server: Apache/2.4.68 ()
- Location: http://glowinglushfinerain.neverssl.com/online/
- Content-Length: 295
- Keep-Alive: timeout=5, max=100
- Connection: Keep-Alive
- Content-Type: text/html; charset=iso-8859-1

The packet bytes pane shows the raw data of the response, including the status line and the location header.

Part 3: Capturing and Analyzing UDP Traffic

Task 1: Generate UDP traffic and capture packets

UDP traffic was generated by streaming a video on YouTube. Since YouTube uses QUIC (a protocol built on top of UDP) for video streaming, this activity produced a continuous flow of UDP packets, which were captured using Wireshark while the video was playing.

UDP Packet Header Details

- **Source Port:** 443
- **Destination Port:** 62484
- **Length:** 1238 bytes
- **Checksum:** 0x7140
- **Observation:** The UDP header is extremely lightweight and simple (consisting of only 4 fields / 8 bytes) compared to the complex TCP header which contains sequence numbers, acknowledgment numbers, window sizes, and flags.

No.	Time	Source	Destination	Protocol	Length	Info
2938	16.964679	2a02:cb81:1003:6::d	2a02:cb80:4029:f0e...	UDP	1292	443 → 62484 Len=1230
2939	16.964680	2a02:cb81:1003:6::d	2a02:cb80:4029:f0e...	UDP	1292	443 → 62484 Len=1230
2940	16.964927	2a02:cb81:1003:6::d	2a02:cb80:4029:f0e...	UDP	1292	443 → 62484 Len=1230
2941	16.964929	2a02:cb81:1003:6::d	2a02:cb80:4029:f0e...	UDP	1292	443 → 62484 Len=1230
2942	16.969190	2a02:cb81:1003:6::d	2a02:cb80:4029:f0e...	UDP	1292	443 → 62484 Len=1230
2943	16.969192	2a02:cb81:1003:6::d	2a02:cb80:4029:f0e...	UDP	1292	443 → 62484 Len=1230
2944	16.969697	2a02:cb81:1003:6::d	2a02:cb80:4029:f0e...	UDP	1292	443 → 62484 Len=1230
2945	16.969697	2a02:cb81:1003:6::d	2a02:cb80:4029:f0e...	UDP	1292	443 → 62484 Len=1230
2946	16.969824	2a02:cb80:4029:f0e...	2a02:cb81:1003:6::d	UDP	94	62484 → 443 Len=32
2947	16.970294	2a02:cb81:1003:6::d	2a02:cb80:4029:f0e...	UDP	1292	443 → 62484 Len=1230
2948	16.970295	2a02:cb81:1003:6::d	2a02:cb80:4029:f0e...	UDP	1292	443 → 62484 Len=1230
2950	16.971553	2a02:cb81:1003:6::d	2a02:cb80:4029:f0e...	UDP	1292	443 → 62484 Len=1230
2951	16.971555	2a02:cb81:1003:6::d	2a02:cb80:4029:f0e...	UDP	1292	443 → 62484 Len=1230
2952	16.971777	2a02:cb81:1003:6::d	2a02:cb80:4029:f0e...	UDP	1292	443 → 62484 Len=1230
2953	16.971778	2a02:cb81:1003:6::d	2a02:cb80:4029:f0e...	UDP	1292	443 → 62484 Len=1230
2954	16.972648	2a02:cb81:1003:6::d	2a02:cb80:4029:f0e...	UDP	1292	443 → 62484 Len=1230
2955	16.972649	2a02:cb81:1003:6::d	2a02:cb80:4029:f0e...	UDP	1292	443 → 62484 Len=1230
2956	16.972889	2a02:cb81:1003:6::d	2a02:cb80:4029:f0e...	UDP	1292	443 → 62484 Len=1230
2957	16.972889	2a02:cb81:1003:6::d	2a02:cb80:4029:f0e...	UDP	1292	443 → 62484 Len=1230
2958	16.972998	2a02:cb80:4029:f0e...	2a02:cb81:1003:6::d	UDP	94	62484 → 443 Len=32
2959	16.976157	2a02:cb81:1003:6::d	2a02:cb80:4029:f0e...	UDP	1292	443 → 62484 Len=1230

> Frame 2943: Packet, 1292 bytes on wire (10336 bits), 1292 bytes captured (10336 bits) on 0

Ethernet II, Src: 02:b0:33:93:70:c0 (02:b0:33:93:70:c0), Dst: Apple-60:27:73 (a4:83:e7:6:

Internet Protocol Version 6, Src: 2a02:cb81:1003:6::d, Dst: 2a02:cb80:4029:f0ed:50f5:7f8

User Datagram Protocol, Src Port: 443, Dst Port: 62484

Source Port: 443

Destination Port: 62484

Length: 1238

Checksum: 0x7140 [unverified]

[Checksum Status: Unverified]

[Stream index: 0]

[Stream Packet Number: 1494]

> [Timestamps]

UDP payload (1230 bytes)

> Data (1230 bytes)

```

0000 a4 83 e7 60 27 73 02 b0 33 93 70 c0 86 dd 60 0a .....s...3-p...
0010 0c 33 04 d6 11 3b 2a 02 cb 81 10 03 00 06 00 00 ...3...;a.....
0020 00 00 00 00 00 0d 2a 02 cb 80 40 29 f0 ed 50 f5 .....*....@)-P
0030 ef 78 f7 5e cd fe 01 bb f4 14 04 d6 71 40 5d 28 ...x...e...q)(
0040 42 21 07 10 43 29 b7 8b a4 38 c9 10 8c 45 f6 8e B!-(...8...E-
0050 8e 49 c7 da 05 0a 65 ec d4 9c 7c db 57 0e 86 b4 I...e...|W
0060 5c 39 90 cc 3d cd 83 e5 48 a1 ab d9 89 42 f0 a0 \9=-...H...B-
0070 68 0f 52 58 97 ed 79 ad d7 f7 11 a6 fc 2b 88 02 hRX-y...+...
0080 ad 9f 87 a3 bb b8 1d a8 00 22 38 f5 c8 e8 68 88 ..... "8...h
0090 f2 20 8a 64 eb fc a3 10 b9 bc c7 00 02 97 22 13 .....d.....
00a0 a4 5e b0 71 09 02 76 09 c6 08 5f 0e 84 2c de e3 ..^q..v.....
00b0 e6 0e b9 2c 98 8f 72 34 a3 7e 9b 43 c0 6e 97 2a .....r4...Cn*
00c0 70 59 69 9c 76 ab b3 f1 df ab b8 da e3 32 ae a7 pY1-v.....2-
00d0 85 0d b2 99 bb 27 ca 0b 48 63 6c aa 8c 9d 5c dd .....Hcl)\
00e0 a3 3e 08 57 27 07 8d 24 08 b0 16 b2 63 12 fb f8 ->-w'-$.U.-&.
00f0 96 9c 8f d6 ab 06 60 80 19 55 ea f5 86 26 f7 20 .....k...U...&.
0100 bc 27 ff 17 ff fd 52 6d 78 4f 00 04 0a be 8c bd .....mRm xO...
0110 6f 30 75 ed ea 60 34 6f dc c2 63 e0 80 ff cf a5 oOu~4o...c...
0120 c2 52 41 28 da bf c4 ca b1 04 3f 95 ed bf e0 83 ..RA(.....?-...
0130 df 47 58 eb c4 8e ca b3 f0 88 24 19 db 3e b4 7c GX-.....?..->]
0140 af 01 27 56 4d 6d 6b c3 d4 44 25 46 26 1b 44 3c 0-Vmmk...D&F&-D<
0150 14 25 f9 ce 14 a3 19 5b f7 2b ce 75 3f 43 17 68 %-[...+u?C-h
0160 5c a9 e7 9f 14 96 4e a0 61 dc d7 93 37 43 fd 5c \...N...a...7C-h
0170 e2 27 f8 ee 4f 7c ca d5 21 5c cb 91 a5 4b 58 36 -[...l^...KX&
0180 e4 23 c4 f6 a0 78 db 53 28 2c 36 95 07 d7 9f ae #-x$S[,6...
0190 f3 70 5a bb e0 b9 a2 93 b6 dc b8 f5 85 0c 48 62 .....pZ.....Hb

```

Part 4: Comparing TCP and UDP by filling in the following tables

Task 1: Fill in the following table and provide reasons.

	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	TCP is connection-oriented; it establishes a connection before transmitting data using the three-way handshake (SYN, SYN-ACK, ACK) and provides reliable transmission via acknowledgments and retransmissions. UDP is connectionless and does not guarantee packet delivery.
Data Integrity and Ordering	TCP	TCP guarantees data integrity and in-order packet delivery using Sequence Numbers and retransmission of lost or corrupted packets. UDP only performs a basic checksum check without reordering or retransmissions.

Task 2: Identify the use Cases and Performance of TCP and UDP.

	TCP	UDP
Use cases	Used when accuracy and completeness are required: • Web browsing (HTTP/HTTPS) • File transfer (FTP/SFTP) • Email (SMTP, IMAP)	Used when speed and low latency are prioritized over reliability: • Streaming (video/ audio) • Online gaming • Voice over IP (VoIP) • DNS queries
Performance	Slower / Higher Overhead due to connection setup, acknowledgments, retransmissions, and congestion control.	Faster / Lower Overhead with a lightweight 8-byte header, no connection setup, and no retransmission delays.